

SIGNAL GENERATOR And INVERTER

Using NE555 Timers

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Often we require a squarewave signal generator with adjustable frequency, almost equal high and low output pulses and adjustable amplitudes. Here we present a simple, useful and low-cost signal generator built around NE555 timers. Using external switches, you can control or select frequency ranges as per your requirements. However, use of frequencies below 30kHz is recommended.

Circuit and working

This project is divided into two parts: Power supply and signal generator.

Power supply. The circuit of the adjustable power supply for the signal generator is shown in Fig. 1. It is built around a step-down transformer (X1), a bridge rectifier (BR1), LM317 adjustable voltage regulator (IC1), two 1N4007 diodes (D1 and D2), two LEDs (LED1 and LED2) and a few other components.

The power supply output available at connector CON2 is variable. You can vary the output from 1.25V to 15V using potentiometer VR1. The variable power supply can be used for

additional adjustment of the timers' amplitude.

Signal generator. The circuit diagram of the signal generator is shown in Fig. 2. It is built around two NE555 timers (IC2 and IC3), an LED (LED3), seven BAT42 Schottky diodes (D3 through D9) and a few other components.

The signal generator produces forward and inverted signals through IC2 and IC3, respectively. Table below shows calculated frequency ranges of squarewave signals.

FREQUENCY RANGES OF SQUAREWAVE SIGNAL GENERATOR

Closed switch	Capacitor	Fmin	Fmax
S1	C1=1nF*	29.7kHz*	357kHz*
S2	C2=10nF	2.97kHz	35.7kHz*
S3	C3=100nF	297Hz	3.57kHz
S4	C4=1μF	29.7Hz	357Hz
S5	C5=10μF	2.97Hz	35.7Hz

*The calculated and actual frequencies may differ significantly above 30kHz due to simplifications of the formula, differences in the parameters of timers and parasitic components in the circuit

IC2 operates as a frequency generator. The frequency (F) is determined by components connected to pins 2, 6 and 7 of IC2 as follows:

$$F = 1 / \{ 0.7(R7 + R8 + 2 \times VR2)C_x \}$$

where C_x may be 1nF, 10nF, 100nF, 1μF or 10μF.

Assuming switch S1 is closed and $C_x = 1nF$ and $VR2 = 22\text{-kilo-ohm}$, the minimal frequency ($F_{\min}$) is given as below:

$$F_{\min} = 1 / \{ 0.7(2k + 2k + 2 \times 22k) 1nF \}$$

$$= 1 / (0.7 \times 48k \times 1nF)$$

$$= 29762Hz$$

$$= 29.7kHz \text{ approx.}$$

When $C_x = 1nF$ and $VR2 = 0\text{-ohm}$, the maximal frequency ($F_{\max}$) is:

$$F_{\max} = 1 / (0.7 \times 4k \times 1nF)$$

$$= 1 / (2800 \times 1nF)$$

$$= 1000000 / 2.8$$

$$= 357142Hz$$

$$= 357kHz \text{ approx.}$$

Here, forward resistances and voltage drops of diodes D3 and D4 are neglected and inverse resistances of diodes assumed to be infinity.

The frequency signal output produced by IC2 is available at connector CON4. The amplitude of signals can be adjusted using potentiometer VR3. Resistive dividers comprising R11 through R14 provide three more amplitudes. That is, the output frequency of IC2 is divided by 10, 100 and 1000. These frequencies are also available across CON4 at pins 3, 4 and 5, respectively.

IC3 works as an inverter. The amplitude of IC3's inverted output available at CON5 can be adjusted using potentiometer VR4. Resistive dividers comprising R17 through R20 provide three more amplitudes by dividing the inverted output by 10, 100 and 1000. These inverted outputs are available across CON5. Diodes D6

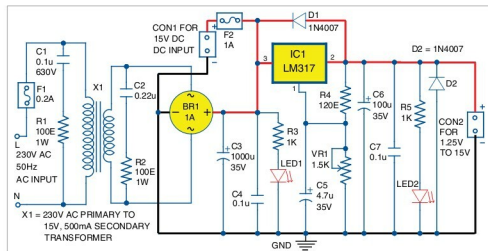


Fig. 1: Power supply circuit